Simple ALM hedging model: Pacific Life RILA, FIA, and IUL crediting segments

Illustrative model. Valuation 2026-10-26, maturity 2027-10-26, T = 1. Insurer sign convention: positive P&L is a gain to the insurer.

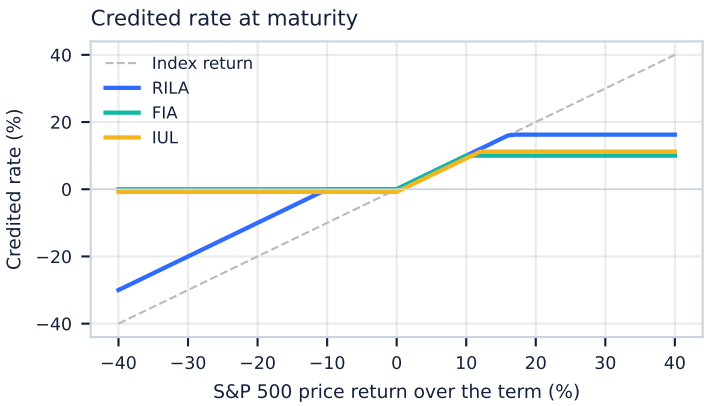
1. Product terms

Three public product designs, each reduced to one 1-year point-to-point crediting segment on the S&P 500 price return (ex dividends), struck on 2026-10-26 and maturing on 2027-10-26. Market as of the valuation date: 1y Treasury 4.47% quoted (continuous r = ln(1 + y) = 4.3730%), dividend yield 1.064%, uniform 1y at-the-money-forward implied volatility 16.72%, forward/spot 1.0336.

Product	Account value	Crediting	Participation	Cap	Protection	Fee
Pacific Protective Growth RILA	$100,000	1-year point-to-point	100%	16.25%	10% buffer	none
Pacific Index Advisory FIA	$100,000	1-year point-to-point	100%	10.00%	0% floor	none
Pacific Horizon IUL 2 (1-Year High Cap Indexed Account)	$10,000	1-year point-to-point	100%	12.00%	0% floor	0.80 pts / yr

2. Cash flows and embedded options

At inception the insurer receives the account value AV_0. At maturity it owes AV_T = AV_0 x (1 + credited rate), where credited rate = participation x option package payoff per $1 - annual deduction. The liability is the PV of AV_T and splits into a bond component (PV of returning AV_0), an option component (PV of the embedded option package), and a fee component (PV of the deduction): liability = bond + option - fee. Insurer value is the negative of the liability. The chart shows the credited rate; the first table shows the resulting account value cash flow at maturity for a few index outcomes; the second lists the option legs per product with their notional and PV.



AV paid at maturity	RILA	FIA	IUL
Index -20%	$90,000	$100,000	$9,920
Index -10%	$100,000	$100,000	$9,920
Index +0%	$100,000	$100,000	$9,920
Index +10%	$110,000	$110,000	$10,920
Index +20%	$116,250	$110,000	$11,120
Credited rate at +20%	16.25%	10.00%	11.20%

Option leg (policyholder view)	Strike (% of start index)	Notional	PV
RILA: Long ATM call (participation)	100.00%	+$100,000	$8,221.22
RILA: Short 1.1625 call (cap)	116.25%	-$100,000	$-2,490.50
RILA: Short 0.90 put (buffer)	90.00%	-$100,000	$-1,761.92
RILA: option component			$3,968.80
FIA: Long ATM call (participation)	100.00%	+$100,000	$8,221.22
FIA: Short 1.10 call (cap)	110.00%	-$100,000	$-4,092.22
FIA: option component			$4,129.00
IUL: Long ATM call (participation)	100.00%	+$10,000	$822.12
IUL: Short 1.12 call (cap)	112.00%	-$10,000	$-350.80
IUL: option component			$471.33

3. Valuation and Greeks

Liability PV per product and in total, with the interest budget (AV_0 - bond PV, what the premium earns at the risk-free rate) and the pricing margin (AV_0 - liability). Greeks are dollar sensitivities of the liability: per 1% index move, per 1%^2, per volatility point, and per basis point of the continuous rate. The insurer's exposure is the negative of each figure.

Liability view ($)	RILA	FIA	IUL	Portfolio
Account value (premium received)	100,000.00	100,000.00	10,000.00	210,000.00
Bond component PV	95,721.26	95,721.26	9,572.13	201,014.65
Option component PV	3,968.80	4,129.00	471.33	8,569.13
Fee component PV	0.00	0.00	76.58	76.58
Liability PV = bond + option - fee	99,690.06	99,850.26	9,966.88	209,507.19
Interest budget = AV - bond	4,278.74	4,278.74	427.87	8,985.35
Pricing margin = AV - liability	309.94	149.74	33.12	492.81
Cap affordable with the full budget	17.84%	10.49%	13.26%	
Delta ($ per 1% index)	518.33	221.99	26.21	766.53
Gamma ($ per 1%^2)	-12.38	0.05	0.09	-12.25
Vega ($ per vol pt)	-207.01	0.76	1.45	-204.80
Rho ($ per bp)	-4.79	-7.77	-0.73	-13.29
Rho of the option component only ($ per bp)	4.79	1.81	0.21	6.81

4. Net exposures

Portfolio exposures from the insurer's side, split by component, and the net option positions that the three products embed once offsetting legs are combined. The insurer is short the liability: it loses when the index rises, is long gamma and long vega (the short cap calls and the RILA buffer put give back more convexity than the long at-the-money calls provide), and its rate exposure sits almost entirely in the bond component.

Insurer view ($)	-Option	-Bond	+Fee	Insurer total
PV ($)	-8,569.13	-201,014.65	76.58	-209,507.19
Delta ($ per 1% index)	-766.53	0.00	0.00	-766.53
Gamma ($ per 1%^2)	12.25	0.00	0.00	12.25
Vega ($ per vol pt)	204.80	0.00	0.00	204.80
Rho ($ per bp)	-6.81	20.10	-0.01	13.29

Net embedded option position (insurer is short this)	Net notional ($)	Policyholder PV ($)	From
Long calls, strike 100.00% of start	+210,000	+17,265	RILA+FIA+IUL
Short calls, strike 110.00% of start	-100,000	-4,092	FIA
Short calls, strike 112.00% of start	-10,000	-351	IUL
Short calls, strike 116.25% of start	-100,000	-2,491	RILA
Short puts, strike 90.00% of start	-100,000	-1,762	RILA
Net embedded option package		**+8,569**	**all**

Unhedged, the option component alone swings by $28,606 across the stress scenarios in section 5; the bond component adds about $1,003 per 50 bp of rates.

5. Hedge recommendations

Instrument	Action	Size	Cost	Purpose
S&P 500 index futures (recommended initial hedge)	Buy	$76,653 exposure	No premium	Offsets $767/1% liability delta
Static alternative: calls at 100.00%	Buy	$210,000 notional	Pay $17,265	Replicates RILA+FIA+IUL
Static alternative: calls at 110.00%	Sell	$100,000 notional	Receive $4,092	Replicates FIA
Static alternative: calls at 112.00%	Sell	$10,000 notional	Receive $351	Replicates IUL
Static alternative: calls at 116.25%	Sell	$100,000 notional	Receive $2,491	Replicates RILA
Static alternative: puts at 90.00%	Sell	$100,000 notional	Receive $1,762	Replicates RILA
Static alternative: net			Pay $8,569 (= option PV)	No rebalancing needed

Futures units: 741.58 at a normalized index level of 100; convert to contracts with the live index level and multiplier. Futures margin, roll, and financing and the dealer spread on the static package are excluded.

Residual risks after the futures (insurer view): gamma +12 per 1%^2, vega +205 per vol point, rho +13 per bp in total (-7 per bp on the options alone). Full-revaluation stress P&L, unhedged and with the futures:

Scenario ($, insurer view)	Bond	Option	Unhedged total	Futures	Option + futures	Hedged total
Index -20%	+0	+16,734	+16,734	-15,331	+1,403	+1,403
Index -10%	+0	+8,120	+8,120	-7,665	+455	+455
Index +10%	+0	-6,900	-6,900	+7,665	+765	+765
Index +20%	+0	-11,873	-11,873	+15,331	+3,458	+3,458
Vol +5 pts	+0	+1,244	+1,244	+0	+1,244	+1,244
Vol -5 pts	+0	-664	-664	+0	-664	-664
Rates +50 bp	+1,003	-338	+664	+384	+46	+1,048
Rates -50 bp	-1,008	+343	-664	-382	-40	-1,047
Selloff: index -15%, vol +8, rates -25 bp	-503	+13,365	+12,863	-11,661	+1,705	+1,202
Range (max - min)		**28,606**	**28,606**		**4,122**	**4,505**

Rebalancing check: after	Liability $/1%	Futures $/1%	Residual $/1%	Residual exposure $	Re-hedge?
Index -10%	759	690	-69	-6,945	yes
Index -5%	775	728	-47	-4,657	yes
Index +5%	728	805	+77	+7,685	yes
Index +10%	660	843	+183	+18,274	yes
Vol +5 pts	685	767	+82	+8,176	yes
Vol -5 pts	910	767	-144	-14,387	yes
One month elapsed, index unchanged	786	764	-21	-2,135	no

Triggers: re-hedge when the residual index exposure exceeds $4,200 (2% of account value), after a volatility move of 5 points or more, at least monthly, and through the final month before 2027-10-26 when the caps and buffer pin delta. Rate risk on the bond component should be managed with general-account asset duration, not with the equity hedge.

6. Explanation of the results

Why the RILA cap is higher but its option package is cheaper. The RILA's 16.25% cap costs $3,969 against $4,129 for the FIA's 10% cap because the policyholder has written the insurer a put struck at 90% of the start index, worth $1,762 on $100,000. That premium funds a cap call struck further out of the money. The FIA and IUL are plain call spreads, so their cost is set entirely by how far out the cap call sits.

Where the money comes from. Each premium dollar earns the risk-free rate for one year; on $210,000 that interest budget is $8,985. The option packages cost $8,569 and the IUL deduction returns $77, leaving $493 of pricing margin before expenses, dealer spreads, and any credit spread earned above Treasuries. Caps are therefore tightly linked to rates and volatility: with the full budget the RILA could afford a 17.84% cap, the FIA 10.49%, and the IUL 13.26%.

What the Greeks say. The liability gains $767 for every 1% rise in the index, almost all of it from the long at-the-money calls. Gamma and vega are negative for the liability ($-12 per 1%^2 and $-205 per vol point), because the short cap calls and the RILA buffer put give back more convexity than the at-the-money calls provide at this volatility. Rho on the options is small; the bond component's rho (about $20 per bp from the insurer's side) dominates and is an asset-duration question.

What the futures hedge does and does not do. Buying $76,653 of index exposure removes the first-order move: an index fall of 20% that would have gained the insurer $16,734 on the options nets to $+1,403 with the futures, and a 20% rise that would have cost $11,873 nets to $+3,458. Both residuals are positive because the insurer is net long gamma, so large moves help the hedged book. The hedge does nothing for volatility: a +5 point vol shock still moves the book by $+1,244, and a vol fall costs money. In the combined selloff the option book finishes at $+1,705 and the total at $+1,202 once the bond component's rate loss of $503 is included.

Rates. A 50 bp rise lowers the PV of the promised account value by $1,003 (a gain for the insurer) and nudges the options and the futures forward. The equity hedge should not be sized to this; it belongs with the general-account assets backing the $201,015 bond component.

Limits of this reading. A single fitted volatility understates the buffer put and overstates the cap calls relative to the observed SPX skew, so the RILA margin is likely optimistic. One year of one segment with no behavior, charges, or hedge costs is a lower bound on complexity, not a reserve or a price.

Data sources

- 1y Treasury constant-maturity yield [observed]: 4.47% (quoted, 2026-10-26). Source: FRED series DGS1 (Board of Governors H.15). https://fred.stlouisfed.org/series/DGS1. Daily series; the value on the valuation date is used as quoted.
- Continuous risk-free rate [fitted/model]: 4.3730% = ln(1 + 0.0447). Source: Derived from DGS1. https://fred.stlouisfed.org/series/DGS1. Treats the quoted yield as annually compounded; bond-equivalent and day-count conventions are ignored.
- S&P 500 dividend yield [observed]: 1.064% (continuous). Source: Robert Shiller, Yale, U.S. stock market data (dividends and prices). http://www.econ.yale.edu/~shiller/data.htm. Trailing dividend yield from the Shiller monthly series around the valuation date; applied as a flat continuous yield.
- 1y at-the-money-forward implied volatility [fitted/model]: 16.72% (uniform across strikes). Source: Fitted from Cboe Option EOD Summary (SPX end-of-day option quotes). https://datashop.cboe.com/option-eod-summary. A single ATMF vol is fitted to the ~1y expiry and applied to every strike; skew and term structure are ignored.
- Volatility cross-check [observed]: VIX level on the valuation date. Source: Cboe VIX historical data. https://www.cboe.com/tradable_products/vix/vix_historical_data/. 30-day index used only as a sanity check on the fitted 1y level, not as a model input.
- Index level [fitted/model]: 100 (normalized). Source: Model convention. Payoffs depend only on the index return, so dollar results do not depend on the level.

Observed inputs are read from the published series on the valuation date. Fitted or model inputs are choices made from observed data: the continuous-rate conversion ignores day count, the single volatility ignores skew and term structure, and the index level is a normalization.

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Things we considered

- One full indexed term for each product, dated 2026-10-26 to 2027-10-26, with the stated cap, buffer or floor, 100% participation, and the IUL's simplified 0.80-point annual deduction.
- Each crediting design decomposed into vanilla S&P 500 option legs per dollar of account value, so the three products price off one model and their risks add.
- The liability split into bond, option, and fee components with an explicit insurer sign convention, so interest budget and pricing margin are visible.
- Closed-form Black-Scholes pricing with analytic delta, gamma, vega, rho, and theta, checked against finite differences in the test suite.
- Observed market inputs (Treasury yield, dividend yield) separated from fitted or model assumptions (continuous-rate conversion, uniform fitted volatility, normalized index level).
- Two hedge constructions: exact static option replication and an initial index-futures delta hedge, with residual gamma, vega, and rho reported.
- Full revaluation under index, volatility, rate, and combined selloff shocks, before and after the futures hedge, with the bond and option pieces shown separately.
- Rebalancing triggers based on the residual index exposure after plausible index, volatility, and time moves.

Things we have not considered

- Other IUL charges: cost of insurance, premium loads, per-policy and per-unit expense charges, rider charges.
- Monthly mechanics: monthly deductions, monthly sweeps into segments, intra-year segment timing.
- Insurance coverage: the IUL death benefit and any mortality or lapse assumptions.
- Premiums: future premium payments and premium-dependent credits or bonuses.
- Withdrawals and surrender: free-withdrawal amounts, surrender charges, and surrender behavior.
- RILA early exit and MVA: interim values, market value adjustments, and performance locks before maturity.
- Optional benefits: riders, guaranteed living or death benefits, and return-of-premium features.
- Loans: IUL policy loans, loan interest, and indexed-loan arbitrage.
- Adviser fees: advisory, platform, or third-party fees deducted from account value.
- Future years: renewal caps, participation rates, and spreads after the first indexed term.
- Real valuation and hedging costs: bid-ask spreads, dealer margins, futures margin and roll costs, collateral, and transaction costs.

Things we can consider in the next steps

- Replace the uniform volatility with the observed SPX strike skew and term structure, which will raise the value of the RILA's buffer put and lower the cap calls.
- Use the full Treasury curve and a dividend futures or forecast curve instead of flat continuous inputs.
- Add the excluded IUL mechanics (cost of insurance, monthly deductions, loans) and the RILA interim value and MVA.
- Introduce policyholder behavior (lapse, withdrawal, surrender) and multi-year renewals with cap resets.
- Price the hedge with real costs: dealer spreads on the option package, futures margin, roll, and financing, and compare the static and dynamic strategies on a cost-adjusted basis.
- Extend to Monte Carlo or a volatility-surface model so that gamma and vega hedges (listed options, variance swaps) can be sized alongside the futures, as the full framework in src/alm_hedging does.
- Back-test the rebalancing rule through the October 2026 selloff path and report realized hedge slippage.